# EXACTNESS OF THE SMALL INVERSE IMAGE AND ITS EFFECT ON STALKS AND REPRESENTABLE SHEAVES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument for all four parts; in part (4) the author gives both the direct computation and the alternative route through the previous parts.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- Small inverse image f^{-1} on sheaves on the small etale site. Same chapter, the functor f^{-1} attached to a morphism of schemes and its description by colimits over etale neighbourhoods. Supplies the functors and identifications appearing in all four parts of the statement.
- Stalks of a sheaf at a geometric point. Same chapter, the stalk functor at a geometric point of the small etale site, and its behaviour under pullback. Supplies parts (2) and (3), including the identification of the stalk of the pullback with the stalk at the composite geometric point.
- Sheaf represented by an etale scheme and its sections. Same chapter, the representable sheaf h_V on Y_etale for V -> Y etale and the computation of f^{-1}h_V. Supplies part (4), namely f^{-1}h_V = h_{X x_Y V}.

*Source reference.* Original lemma 0034: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4658–4658. Navigation pointer invoked at proof line 4658; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00X6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4640–4640. Navigation pointer invoked at proof

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line 4640; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original equation 04I1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4432–4432. Navigation pointer invoked at proof line 4641; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 00YR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4680–4680. Navigation pointer invoked at proof line 4680; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 04JC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4682–4682. Navigation pointer invoked at proof line 4682; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 4684; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 05V1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4688–4688. Navigation pointer invoked at proof line 4688; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 03PT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3456–3456. Navigation pointer invoked at proof line 4690; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original equation 04I3: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4632–4632. Navigation pointer invoked at proof line 4697; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 001P: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4713–4713. Navigation pointer invoked at proof line 4713; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 04D3: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4721–4721. Navigation pointer invoked at proof line 4721; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $f : X \to Y$ be a morphism of schemes.*

(1) *The functor $f^{-1} : Ab(Y_{étale}) \to Ab(X_{étale})$ is exact.*
(2) *The functor $f^{-1} : Sh(Y_{étale}) \to Sh(X_{étale})$ is exact, i.e., it commutes with finite limits and colimits, see Categories, Definition 0034.*
(3) *Let $\overline{x} \to X$ be a geometric point. Let $\mathcal{G}$ be a sheaf on $Y_{étale}$. Then there is a canonical identification*
$$(f^{-1}\mathcal{G})_{\overline{x}} = \mathcal{G}_{\overline{y}}.$$
*where $\overline{y} = f \circ \overline{x}$.*
(4) *For any $V \to Y$ étale we have $f^{-1}h_V = h_{X \times_Y V}$.*

*Proof.* The exactness of $f^{-1}$ on sheaves of sets is a consequence of Sites, Proposition 00X6 applied to our functor $u$ of Equation (04I1). In fact the exactness of pullback is part of the definition of a morphism of topoi (or sites if you like). Thus we see (2) holds. It implies part (1) since given an abelian sheaf $\mathcal{G}$ on $Y_{étale}$ the underlying sheaf of sets of $f^{-1}\mathcal{F}$ is the same as $f^{-1}$ of the underlying sheaf of sets of $\mathcal{F}$, see Sites, Section 00YR. See also Modules on Sites, Lemma 04JC. In the literature (1) and (2) are sometimes deduced from (3) via Theorem 03PU.

Part (3) is a general fact about stalks of pullbacks, see Sites, Lemma 05V1. We will also prove (3) directly as follows. Note that by Lemma 03PT taking stalks commutes with sheafification. Now recall that $f^{-1}\mathcal{G}$ is the sheaf associated to the presheaf
$$U \longrightarrow \operatorname{colim}_{U \to X \times_Y V} \mathcal{G}(V),$$
see Equation (04I3). Thus we have
$$\begin{aligned}(f^{-1}\mathcal{G})_{\overline{x}} &= \operatorname{colim}_{(U,\overline{u})} f^{-1}\mathcal{G}(U) \\ &= \operatorname{colim}_{(U,\overline{u})} \operatorname{colim}_{a:U \to X \times_Y V} \mathcal{G}(V) \\ &= \operatorname{colim}_{(V,\overline{v})} \mathcal{G}(V) \\ &= \mathcal{G}_{\overline{y}}\end{aligned}$$
in the third equality the pair $(U, \overline{u})$ and the map $a : U \to X \times_Y V$ corresponds to the pair $(V, a \circ \overline{u})$.

Part (4) can be proved in a similar manner by identifying the colimits which define $f^{-1}h_V$. Or you can use Yoneda's lemma (Categories, Lemma 001P) and the functorial equalities
$$\operatorname{Mor}_{Sh(X_{étale})}(f^{-1}h_V, \mathcal{F}) = \operatorname{Mor}_{Sh(Y_{étale})}(h_V, f_*\mathcal{F}) = f_*\mathcal{F}(V) = \mathcal{F}(X \times_Y V)$$
combined with the fact that representable presheaves are sheaves. See also Sites, Lemma 04D3 for a completely general result. □